

gr00t-N1.5 — Outlier magnitude heatmap ($\log_{10}(|x|/\text{scale})$ under per-block MSE-optimal int4 scaling)
Blue = crushed to int4 level 0; white $\approx$ 1 int4 level; red = saturating ceiling ($\log_{10}(7.5)\approx 0.87$)

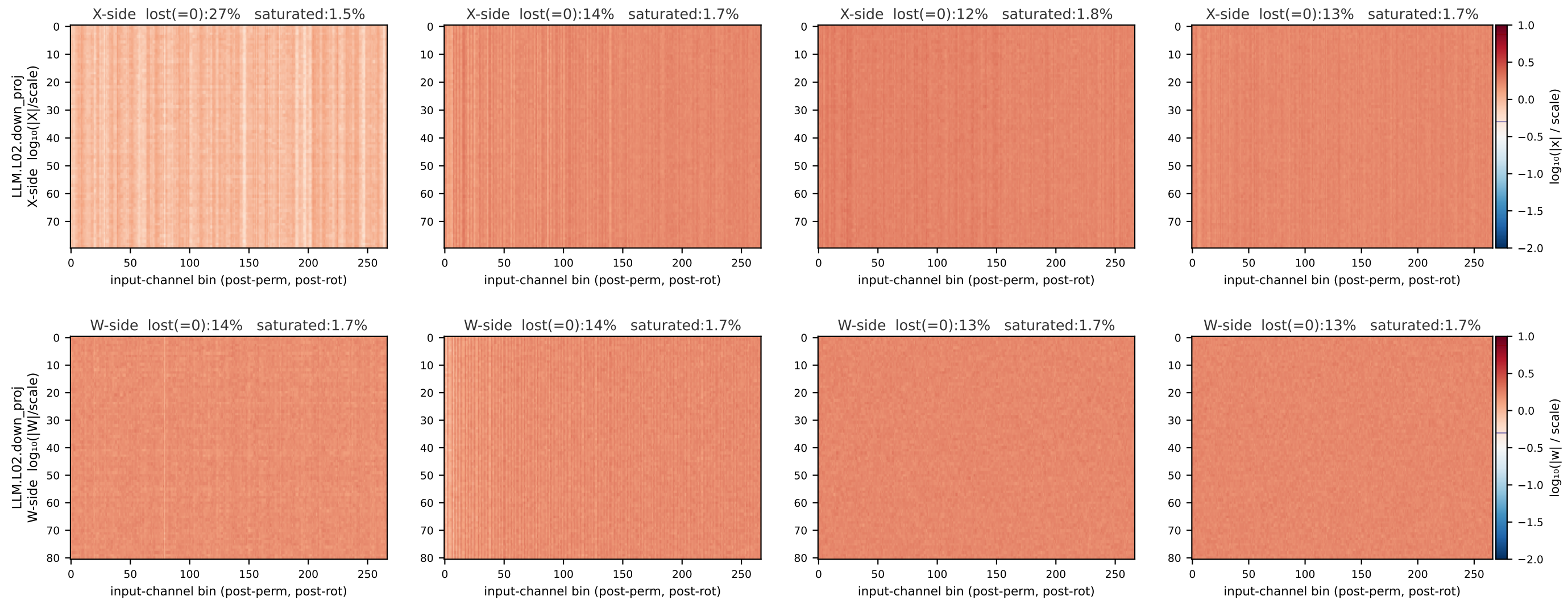
raw (no rotation)

DuQuant SVD

Hadamard only

SVD-Hadamard (A2-lite)

LLM.L02.down_proj — structural pathology (raw amax-ratio 48000 $\times$)



LLM.L02.gate_proj — typical MLP layer (SVD-only barely moves amax-ratio)

